



PAEFF: Precise Alignment and Enhanced Gated Feature Fusion for Face-Voice Association

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Abstract

We study the task of learning association between faces and voices, which is gaining interest in the multimodal community lately. These methods suffer from the deliberate crafting of negative mining procedures as well as the reliance on the distant margin parameter. These issues are addressed by learning a joint embedding space in which orthogonality constraints are applied to the fused embeddings of faces and voices. However, embedding spaces of faces and voices possess different characteristics and require spaces to be aligned before fusing them. To this end, we propose a method that accurately aligns the embedding spaces and fuses them with an enhanced gated fusion thereby improving the performance of face-voice association. Extensive experiments on the VoxCeleb dataset reveals the merits of the proposed approach¹.

Index Terms: Multimodal learning, Face-voice association, Cross-modal verification & matching, Hyperbolic Space.

1. Introduction

Face-voice association is a widely studied task in cognitive science, which investigates the relationship of the human faces with their voices [1, 2, 3]. This task is brought into the computer vision community by Nagrani et al. [4] by formulating a deep learning approach to identify which face pairs belong to the voice segment. Consequently, several methods [5, 6, 7, 8, 9, 10, 11, 12, 13] have been proposed to tackle this problem. These methods rely on unimodal encoders to separately extract the face and voice embeddings followed by learning the discriminative space which minimizes the distance between the embeddings of same identities. Typically, contrastive or triplet loss formulations are leveraged to achieve this task [5, 6, 4, 14, 15]. Although demonstrating favorable performance in learning discriminative space, such loss formulations are limiting in the sense that they require tuning of a margin hyperparameter which is challenging as the distances between samples can differ significantly during training. Furthermore, the training complexities of contrastive and triplet losses are $O(n^2)$ and $O(n^3)$, respectively (where n is the number of samples for a face or voice) resulting in longer training time as the size of the dataset increases dramatically. To this end, several methods have explored the utilization of auxiliary identity information as an alternative to contrastive or triplet loss formulations. For example, Nawaz et al. [8] introduced an approach to

learn the discriminative embedding space by leveraging identity centers, eliminating the need of pairwise or triplet supervision. These methods are fairly effective, and require the necessity of being utilized with traditional classification losses such as cross-entropy (CE) loss, resulting in counterintuitive combination of Euclidean and angular spaces respectively. To mitigate this, Saeed et al. [10] have proposed the fusion and orthogonal projection (FOP) method that introduces a loss by imposing orthogonality constraints on the fused embeddings of faces and voices to learn discriminative embedding space. FOP performs favorably on the face-voice association task, however, fusing the features of two modalities without aligning the corresponding embedding spaces restricts the performance.

In this work, we hypothesize that the fusion of feature representations that come from different embedding spaces is ineffective unless the two embedding spaces are aligned, and it limits the ability of loss formulation to maximize the distance between the inter-identity representations. Therefore, a mechanism is required that can align the embedding spaces of two modalities before fusing them. To this end, we propose a dual branch face-voice association framework that enables the model to align the embeddings of faces and voices in hyperbolic space by preserving the distances and complex relationships before fusing them. In addition, we propose a fusion module that can effectively fuse the feature representations resulting in performance improvement of face-voice association tasks. In summary, our contributions are as follows.

- We propose a dual branch face-voice association framework that precisely aligns the multimodal feature embeddings before merging them.
- We introduce an effective feature fusion strategy that utilizes the attention weights to merge face and voice feature embeddings, thereby improving model performance.
- We perform extensive experiments on a challenging and benchmark face-voice association dataset (VoxCeleb [16]) to demonstrate the effectiveness of the proposed framework.

2. Background

This section summarizes previous work related to multimodal learning, face-voice association and hyperbolic embedding space.

Multimodal learning and face-voice association. Multimodal learning has been an important research area in recent years due to the fact that information in real-world scenarios comes through multiple and diverse data sources [17, 18, 19, 20, 21].

¹Our code is available at: <https://github.com/hannabdul/paeff>

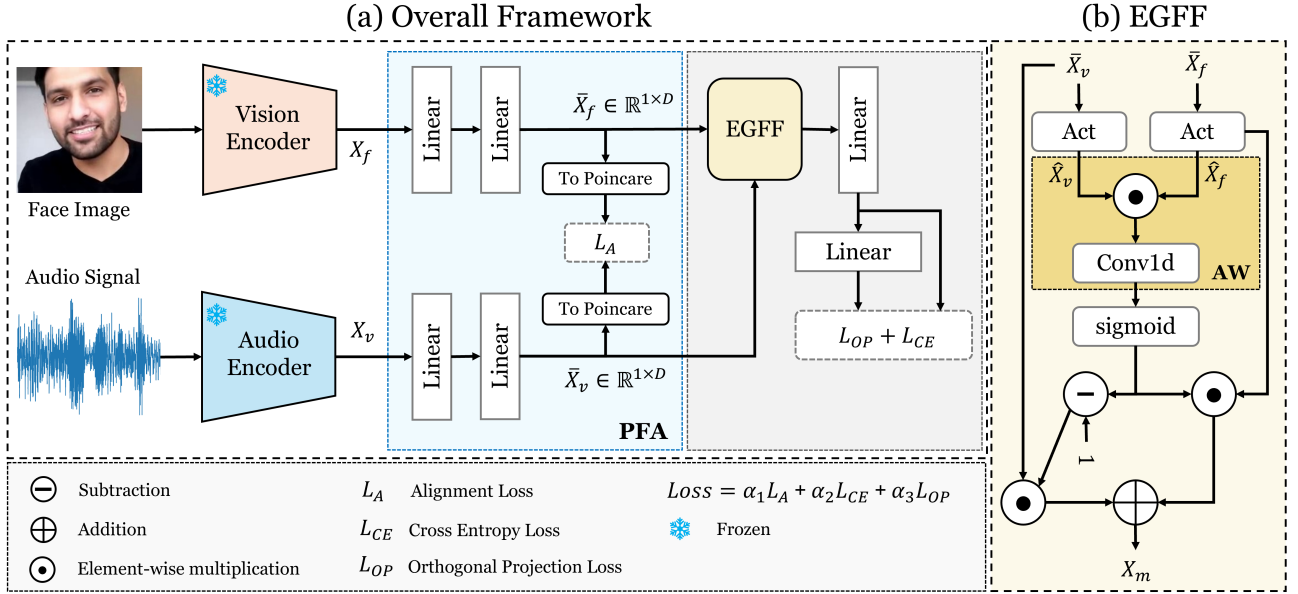


Figure 1: (a) Overall illustration of the proposed face-voice association approach. Face (X_f) and voice (X_v) features are extracted by utilizing vision and audio encoders, respectively. Extracted features X_f and X_v are then fed to linear layers to obtain the projected features of dimension D . The projected features are transformed to hyperbolic space ($\mathbb{H}_2$) for accurate alignment of feature representations. Symmetric cross-entropy loss L_A is utilized to align the feature representations. Afterwards, aligned features are fused by using enhanced gated feature fusion (EGFF) module. The fused features are fed to the logits layer and optimized by means of orthogonal projection (L_{OP}) and cross-entropy (L_{CE}) losses. (b) Detailed overview of EGFF is shown on the right side. AW refers to the computation of attention weights for feature fusion.

For instance, human behaviors, emotions, actions, and humor consist of multiple modalities such as speech, spoken language, and faces, thus various human-centered tasks are extensively studied including face-voice association [6]. Nagrani et al. [4, 5] illustrated that the distance between face and voice embeddings can be minimized by using either contrastive or triplet loss formulation to learn face-voice association. Later, Nawaz et al. [8] proposed a method to learn a discriminative embedding space by utilizing identity centers, removing the requirement for pairwise or triplet supervision. More recently, Saeed et al. [10] utilized the fused embeddings and employed the orthogonality constraints to minimize the distance between the feature embeddings of same classes. Since face and voice feature representations belong to different embedding spaces, therefore, we propose to align these embeddings to obtain discriminative embedding space.

Hyperbolic Space. The aim of embedding a space into another requires the preservation of complex relationships of the feature representations. Sala et al. [22] showed that hyperbolic space can better preserve distances and complex relationships than Euclidean space. The authors explained how hyperbolic spaces are related to Poincare disk $\mathbb{H}_2$ that contains all the points in a unit disk in two dimensions. Later, Khrulkov [23] showed that hyperbolic embeddings are suitable for various computer vision tasks including image classification, image retrieval, and few-shot learning. More recently, hyperbolic space is utilized for various multimodal tasks [24, 25, 26]. Building on this insight, we utilized hyperbolic space to effectively model semantic similarities between faces and voices, enhancing the performance of face-voice association task.

3. Method

3.1. Baseline Approach

In this work, we adapt a two-branch framework as our baseline method to establish face-voice association [10]. The baseline method utilizes the pre-trained networks to extract features of faces and voices. Afterwards, an attention-based fusion module is used to merge the feature embeddings. These fused embeddings encapsulate the semantics of the identities and orthogonality constraint is applied to maximize the separability between the different identities while minimizing the intra-identity separation. This utilization of orthogonality on fused embeddings provides fair improvement for face-voice association task. However, feature representations of faces and voices belong to different embedding spaces. Therefore, it is crucial to align the embedding spaces of faces and voices before merging them. To this end, we propose an approach that precisely aligns the feature embeddings in hyperbolic space before fusing them, thereby improving performance of the model. We describe our approach in detail in the next section.

3.2. Overall Architecture

The overall architecture of the proposed framework is illustrated in Figure 1. The proposed framework extracts the features X_f and X_v of the face and voice inputs, respectively, by using the pre-trained face and voice encoders. Following that, we utilize two linear layers to project the feature embeddings to the same dimension. As discussed earlier, feature representations of different modality spaces need to be precisely aligned before feature fusion. This alignment encourages the model to put emphasis on the features of similar identities during fusion while com-

Table 1: *Cross-modal verification results on seen-heard and unseen-unheard configurations of the proposed method and existing state-of-the-art methods. Best results are highlighted in bold text whereas the second best are underlined.*

Method	EER ↓	AUC ↑	EER ↓	AUC ↑
	Seen-Heard		Unseen-Unheard	
DIMNet [7]	-	-	24.9	-
Learnable Pins [5]	21.4	87.0	29.6	78.5
MAV-Celeb [14]	-	-	29.0	78.9
Single Stream Net. [8]	17.2	91.1	29.5	78.8
Multi-view [28]	-	-	28.0	-
AML [29]	-	<u>92.3</u>	-	80.6
DRL [30]	-	-	25.0	84.6
FOP [10]	19.3	89.3	24.9	83.5
SBNNet. [11]	-	-	25.7	82.4
PAEFF (Ours)	14.3	93.8	22.9	<u>84.4</u>

plementing the orthogonality constraint to maximize the separability between the different identities. To this end, we make face-voice pairs within each batch and minimize the cosine similarity scores between the irrelevant pairs while maximizing the similarity of the relevant face-voice pairs [27]. We utilize symmetric cross-entropy loss $\mathcal{L}_A$ to learn the feature alignment between the modality spaces. Moreover, noticing that the hyperbolic space $\mathbb{H}_2$ can preserve the distance and complex relationships better as compared to Euclidean space [22], we first project the feature embeddings via Poincaré to hyperbolic space before performing feature alignment. This process encourages the model to effectively align the feature spaces. Later on, we fuse the aligned features by using enhanced gated feature fusion (EGFF). The fused features are then fed to a linear layer to obtain projected feature embeddings. Finally, a classification layer is used to project the features for obtaining the prediction logits. Following the prior work [10], we utilize the orthogonal projection loss $\mathcal{L}_{OP}$ and cross entropy loss $\mathcal{L}_{CE}$ for training the model. The total loss $\mathcal{L}$ is the combination of alignment, orthogonal projection, and cross entropy losses given as:

$$\mathcal{L} = \alpha_1 \mathcal{L}_A + \alpha_2 \mathcal{L}_{OP} + \alpha_3 \mathcal{L}_{CE} \quad (1)$$

where $\mathcal{L}_A$ is the precise feature alignment loss, $\mathcal{L}_{OP}$ is the orthogonal projection loss, $\mathcal{L}_{CE}$ cross-entropy loss, and $(\alpha_1, \alpha_2, \alpha_3)$ are the hyperparameters.

3.3. Enhanced Gated Feature Fusion

Earlier work [10] demonstrated that gated feature fusion [31] provides better performance compared to the linear fusion strategy for the face-voice association task. We therefore selected the gated fusion strategy for fusing the features of two modalities as shown in Figure 1. Given the aligned feature representations $\hat{X}_f$ and $\hat{X}_v$ of the face and voice modalities, we first apply the non-linear activation to the feature representations to obtain features $\hat{X}_f$ and $\hat{X}_v$, respectively. Then, element-wise multiplication is performed between $\hat{X}_f$ and $\hat{X}_v$ followed by convolution layer and *sigmoid* activation to obtain the attention features X_a . The element-wise multiplication on the aligned features is pivotal as it highlights the significantly aligned feature representations. Finally, the fused features are obtained by using the equations below:

$$\begin{aligned} X_a &= \sigma(\text{conv}(\hat{X}_f \odot \hat{X}_v)) \\ X_m &= X_a \odot \hat{X}_f + (1 - X_a) \odot \hat{X}_v \end{aligned} \quad (2)$$

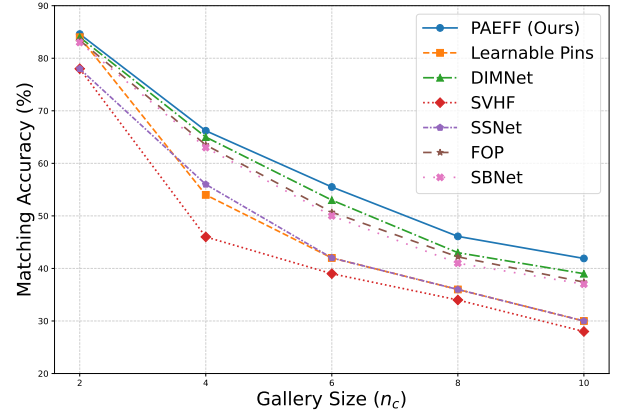


Figure 2: *Cross-modal matching results of the proposed model and existing SOTA methods with varying gallery size (n_c).*

where σ is the sigmoid operation and $\odot$ represents the element-wise multiplication between the operands.

4. Experiments

Implementation Details: In our experiments, we utilize one Quadro RTX 6000 GPU to train the proposed model for 50 epochs using a batch-size of 1024. We set the initial learning rate to $2e^{-5}$. During training, we utilize the AdamW optimizer with cosine scheduler. We empirically set the hyperparameters α_1 , α_2 , and α_3 to 0.3, 0.35, and 0.35 respectively. Following prior work [10], we employed VGGFace [32] and Utterance Level Aggregation [33] encoders to extract face and voice feature embeddings.

Dataset: We perform experiments with VoxCeleb¹, a benchmark dataset to establish face-voice association. VoxCeleb is an audio-visual dataset consisting of short clips of human speech, extracted from interview videos uploaded to YouTube [16]. The dataset provides over 100,000 speaking face-tracks from over 20,000 videos belonging to 1,251 speakers. Following prior work [5, 8, 10, 11], we used the same train, validation, and test splits to evaluate on seen-heard and unseen-unheard configurations. The seen-heard split consists of disjoint videos from the same set of speakers while the unheard-unseen split contains disjoint identities. Face-voice association is typically established using cross-modal verification and cross-modal matching tasks. In cross-modal verification, the goal is to determine whether a given audio segment and a face image correspond to the same identity, while cross-modal matching involves comparing the input modality (probe) to other modality with the varying gallery size n_c .

Evaluation Metrics: Following prior works [5, 10, 11], we utilize the *Equal Error Rate* (EER), *Area Under the Curve* (AUC), and *Matching Accuracy* metrics to evaluate the performance on cross-modal verification and matching tasks.

4.1. Face-Voice Association Results

Cross-modal verification: We compare the performance of the proposed method against several state-of-the-art (SOTA) face-voice association methods including DIMNet [7], Learnable Pins [5], MAV-Celeb [14], Single Stream Network [8], Multi-view [28], Adversarial Metric learning [29] (AML), Disentangled Representation Learning [30] (DRL), FOP [10], and

¹<https://www.robots.ox.ac.uk/~vgg/data/voxceleb/>

Table 2: Cross-modal verification results (AUC%) under varying demographics for seen-heard and unseen-unheard configurations. Following prior work [5], rand. configuration is obtained without imposing any demographic restriction. In contrast, G, N, A, GNA are derived by applying their respective demographic constraints. †Values are taken from prior work [10]. Best results are highlighted in bold text.

Method†	Demographic									
	Rand.	G	N	A	GNA	Rand.	G	N	A	GNA
	Seen-Heard					Unseen-Unheard				
CE	86.6	78.0	85.0	86.3	77.3	81.7	65.9	53.6	76.0	52.8
Center	88.6	79.2	87.0	88.2	78.1	77.5	62.4	51.7	72.5	54.2
Git	88.9	79.7	87.4	88.6	78.5	77.9	62.6	51.8	72.8	54.2
Contrastive	84.7	69.7	83.7	84.5	69.2	79.5	61.0	53.5	74.7	51.8
Triplet	88.0	76.3	86.7	87.6	75.6	81.7	65.5	53.4	76.3	52.2
FOP	89.3	76.7	87.9	88.6	76.6	83.5	68.8	54.9	78.1	54.2
PAEFF (Ours)	93.8	87.1	92.3	93.4	86.3	84.4	71.5	53.0	78.8	54.0

Single Branch Network [11] (SBNet). Table 1 shows the results of cross-modal verification task for both seen-heard and unseen-unheard splits. We observe that the proposed method outperforms the other SOTA methods in terms of EER metric by achieving the EER scores of 22.9% and 14.3%, respectively, for unseen-unheard and seen-heard splits whereas the existing best FOP [10] obtains the EER score of 24.9% and 19.3%, respectively. In terms of AUC metric, our method obtains the favorably better score of 93.8% as compared to the AML [29] that achieves the score of 92.3% on seen-heard split. However, the proposed method provides on-par performance for unseen-unheard configuration and achieves the AUC score of 84.4%.

Cross-modal matching: In addition to the cross-modal verification task, we compare the performance of the proposed model with SOTA methods for face-voice matching task in Figure 2. It is evident from the results that the proposed model has better matching performance (shown in blue color) as compared to the SOTA method DIMNet [7] (highlighted in green color) for all gallery sizes (n_c where c in [2, 4, 6, 8, 10]). The proposed method achieves an accuracy of 84.6% for gallery size 2 and 41.9% for gallery size 10. Notably, our method provides significantly better matching performance when the gallery size increases to 8 or 10 whereas performance of the other methods degrades considerably for the higher gallery size.

Comparison on varying demographic combinations: Table 2 presents the effect of Gender (G), Nationality (N), Age (A), and their combination (GNA) separately on face-voice association problem. We notice that the proposed method consistently outperforms the other approaches in terms of AUC score for the seen-heard split of all demographics achieving a considerable gain of 7.8% for GNA combination. For unseen-unheard configuration, the proposed approach performs favorably for the random, G, and A combinations by obtaining AUC scores of 84.4%, 71.5%, and 78.8%, respectively. However, we observe that the FOP, Center, and Git loss approaches provide better performance on N and GNA combinations of unseen-unheard split.

4.2. Ablation Study

We demonstrate the effectiveness of each of the proposed components in Table 3. For the unseen-unheard configuration, the baseline approach achieves EER and AUC scores of 27.6% and 79.2%, respectively. Replacing the gated fusion mechanism in the baseline method with the proposed EGFF module provides an AUC gain of 1.5% and reduces the EER score by 1.6%. Similarly, aligning the embedding spaces of two modalities before feature fusion leads to significant improvement in the AUC and EER metrics achieving scores of 83.6% and 23.9%, respec-

Table 3: Ablation study demonstrating the effectiveness of proposed components on both seen-heard and unseen-unheard configurations. FA represents the feature alignment in Euclidean space before fusion. Best results are highlighted in bold text.

Method	EER ↓	AUC ↑	EER ↓	AUC ↑
	Seen-Heard		Unseen-Unheard	
Baseline	26.5	81.0	27.6	79.2
Baseline + EGFF	23.8	84.1	26.0	80.7
Baseline + EGFF + FA	14.9	92.1	23.9	83.6
PAEFF (Ours)	14.3	93.8	22.9	84.4

Table 4: Ablation of computing the attention weights for fusion of face and voice features on unseen-unheard configuration. Best results are highlighted in bold text.

Method	EER ↓	AUC ↑
Addition	24.4	84.4
Concatenation	24.0	84.3
Multiplication	22.9	84.4

tively. Finally, projecting the feature representations to the hyperbolic space and aligning the embedding spaces further improves the face-voice association performance and achieves the state-of-the-art results of 22.9% in terms of EER metrics, underscoring the merits of the proposed contributions.

Table 3 also reports the performance comparison of the proposed components on seen-heard split. We observe the similar performance improvement when the proposed components are integrated in the baseline approach. In particular, we notice that there is a significant reduction of 8.9% in EER score and considerable gain of 8.0% in AUC when the embedding spaces of the two modalities are aligned before feature fusion. However, the reduction in EER score is not considerable when feature representations are aligned in the hyperbolic space in case of seen-heard configuration.

Computation of Attention Weights: The attention weights utilized to highlight the relevant semantic features in EGFF (Figure 1) are crucial. Therefore, we investigate how to combine face and voice features for obtaining rich semantic information. Table 4 presents the results of obtaining attention weights by utilizing addition, concatenation, and multiplication operations for feature fusion. We observe that the AUC score of the proposed approach is not affected by the type of operation used for computing attention weights. However, the multiplication operation is better for computing attention weights when face and voice features are fused using gated feature fusion mechanism, improving EER score.

5. Conclusion

In this work, we introduced an effective approach that accurately aligns the face-voice embeddings before fusion. We demonstrated that the precise alignment of features is a crucial step for obtaining better performance in the face-voice association task. Moreover, we showed that effectively fusing the face-voice feature representations also improves the results of the face-voice association task. Extensive experiments conducted on the VoxCeleb benchmark reveal the merits of the proposed approach. Our potential future direction is to explore the adaptation of the proposed approach for multilingual face-voice association and speaker diarization.

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